

Airbnb Listing Clone — As-Built Architecture

What was actually built, and how the reference was measured rather than copied · desktop-only, 1910 × 1000, DPR 1

THREE GRADED VIEWS

One route. The two overlays are URL state, not component state — so they are deep-linkable and survive a reload, which is how the reference behaves.

Listing Page
/ — header, hero gallery, booking card, 9 content sections. RSC-rendered.

Photo Tour
/?modal=PHOTO_TOUR_SCROLLABLE — 9 sections, 43 photos, [lead, pair*] repeating layout.

Lightbox
&idx=n — arrows, ←/→ keys, focus trap, two-level dismissal.

APPLICATION — NEXT.JS 16 APP ROUTER

React 19.1 · TypeScript 5.9 · Turbopack. Server Components by default; client boundaries only where interaction demands one.

Server Components
Listing data resolved on the server. No client fetch for first paint.

Overlay router
Search params drive both overlays. Back button steps the stack.

Focus management
Trap on open, restore to the invoking control on close.

next/font/local
Airbnb Cereal VF — one variable file, weights 200–900.

next/image
73 captured assets, AVIF/WebP, explicit sizes to avoid CLS.

DESIGN SYSTEM — TAILWIND V4 CSS-FIRST

Every visual value is a token in @theme. A hardcoded hex fails check:tokens in CI, so the scale cannot silently drift.

Type scale
Single 1.43 line-height ratio, measured across 292 nodes. 11 roles deliberately deviate.

Colour
14 backgrounds, 7 foregrounds, 8 border colours — all measured, none guessed.

Elevation & radius
5 shadows, 11 radii, read from the reference's computed styles.

Motion
Durations and easings tokenised; transitions match the reference's timings.

REFERENCE MEASUREMENT PIPELINE

The reference actively defends itself. This tier exists because the obvious approaches all fail — and because the brief forbids lifting the code.

CDP attach
Connect to a browser a human started. A Playwright-launched one is 429'd by BotID.
THE ONLY CHANNEL

Native getComputedStyle
The page stubs it to return an empty declaration. Recovered off a blank iframe's contentWindow.
DEFENCE BYPASSED

Geometry capture
getBoundingClientRect on every node. Marked EXACT or APPROX — never unlabelled.

Asset capture
73 files / 11.71 MB, zipped in-page by the page's own JS and downloaded once.

Why not just copy?
The brief states a plagiarism-detection mechanism is in place and that lifting the codebase can mean disqualification. getComputedStyle being neutered means colours, spacing, type scale and timings cannot be scraped at all. The page is designed to be measured and rebuilt — so geometry and text anchor the layout, and every visual value is re-derived and written down with its confidence.

VERIFICATION — NPM RUN VERIFY

Gates run in CI order. The last one exists because all the others compare numbers, and numbers were passing while the human could see differences.

tsc + lint
Zero errors enforced.

check:tokens
Fails on any hardcoded visual value.

check:copy
Reads the RENDERED page, not the source module.

check:photos
43 photos, 9 sections, hero slot order.

verify-overlays
32 behavioural assertions — keyboard, focus, URL, counters.

check:visual
Pixel diff vs the reference at a locked viewport.
PIXELS, NOT NUMBERS

BACKEND — BUILT, THEN PAUSED BY SCOPE DECISION

Phase 2 made the controls real. Later de-scoped to concentrate the remaining time on visual parity, which is what the brief grades.

Route Handlers
5 booking routes.
PAUSED

Prisma + SQLite
Seeded DB committed — zero reviewer setup.
PAUSED

Server-side pricing
Total never read from the request body.
PAUSED

Availability re-check
Inside the write transaction.
PAUSED

verify-booking
Posts a tampered total, then double-books. Expects the real price and a 409.
PAUSED

DECISIONS WORTH DEFENDING

1 · Overlays are URL state
Both overlays live in search params, not useState. That is what makes them deep-linkable, reload-safe and Back-navigable — matching behaviour, not just appearance. The brief grades behavioural parity explicitly.

2 · Measure the rendering, never the source
The copy checker reads the rendered page. Reading strings out of lib/listing.ts would mean re-implementing JS escape semantics — which is the exact layer the bug hides in.

3 · Numbers or PENDING, never prose
"Rounded corners" with no value is not a specification, it is an invitation to guess. One such gap produced an 18px radius where the reference uses 12px on a single clipping wrapper.

4 · A gate that can skip is worse than no gate
The copy checker's first version went green whenever it could not reach a page — inside a script that never starts a server. It would have read as coverage forever. Exactly one skip condition is now allowed, and it is explicit.

5 · Fonts are unobservable
fontFamily returns the declared stack, never the resolved face, and document.fonts.check() returns true for a font that never loaded. A whole pass of widths was taken against Segoe metrics before the tell was found: height exact + width wrong = glyphs, not CSS.

6 · Disclosure is a holding position
While the channel was blocked, invented copy was isolated in one module and disclosed. That constraint is what later made replacing it with measured strings a data edit rather than a rewrite.

7 · Shared state is not just files
Single-writer file ownership broke three times — on a shared type, a shared .next directory, and the shared git index. Whoever changes a shared thing owns every consumer of it in the same commit.

3 views
listing · tour · lightbox

43 photos
9 tour sections

32
behavioural assertions

Red outline marks a defence that had to be defeated, or the one gate that compares pixels · grey marks de-scoped work

Sheet 2 — as-built · see Sheet 1 for the production scaling architecture